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20-year Global Trend Evidence on the Innovation Structure of Surgical Robotics Across the Technology-to-Clinic Continuum --Manuscript Draft--

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Corresponding Author:	Kai Huang Xiangya Hospital Central South University Department of Dermatology CHINA
First Author:	Jingyi Dong
Order of Authors:	Jingyi Dong
	Kai Sun
	Haolin Chen
	Jingyi Zeng
	Xiangxin Zhang
	Yifu Xia
	Wangda Guo
	Ying Wang
	Zexu Lin
	Xingning Xie
	Bojian Liu
	Kai Huang
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Authors:

Jingyi Dong^{1, 2, 3, 4, 5, †}, Kai Sun^{1, 2, 3, 4, 5, †}, Haolin Chen⁶, Jingyi Zeng⁷, Xiangxin Zhang⁷, Yifu Xia^{1, 2, 3, 4, 5}, Wangda Guo^{1, 2, 3, 4, 5}, Ying Wang^{1, 2, 3, 4, 5}, Zexu Lin^{1, 2, 3, 4, 5}, Xingning Xie^{1, 2, 3, 4, 5}, Bojian Liu^{1, 2, 3, 4, 5}, Kai Huang^{1, 2, 3, 4, 5, *},

Affiliations:

¹ Department of Dermatology, Xiangya Hospital, Central South University, Hunan, 410008, China.

² Furong lab, Central South University, Hunan, 410008, China.

³ National Engineering Research Center of Personalized Diagnostic and Therapeutic Technology, Xiangya Hospital, Central South University, Hunan, 410008, China.

⁴ Hunan Key Laboratory of Skin Cancer and Psoriasis, Xiangya Hospital, Central South University, Hunan, 410008, China.

⁵ National Clinical Research Center for Geriatric Disorders, Xiangya Hospital, Hunan, 410008, China.

⁶ School of Computer Science and Engineering, Central South University, Hunan, 410012, China.

⁷ Xiangya School of Public Health, Central South University, Hunan, 410008, China.

*Corresponding author. Email: kaiserhuang@csu.edu.cn

†These authors contributed equally to this work.

Abstract:

Background: Technological innovation in surgical robotics often surface long before they are clinically applied. But only a select number of patented designs are successfully developed into approved surgical systems. Early on, numerous engineering decisions are secured by patents, and these choices will shape subsequent technological paths and their ability to reach clinical trials. However, empirical evidence linking upstream technology choices to downstream regulatory outcomes remains very limited.

Methods: We first constructed and curated a global patent corpus, with 39,629 patents from 102 countries and curated 30,016 regulatory data of robot-assisted medical devices and products from

the public databases of the US Food and Drug Administration and China's National Medical Products Administration. We then linked patents to regulatory-cleared products using assignee identity, temporal consistency, and technical correspondence, labeling patents as translated or non-translated and stratifying translated patents into four levels (Q1–Q4) by translational intensity. Next, we analyzed temporal, geographic, legal, and technical characteristics of the patent landscape and applied topic modeling to compare patent activity with patterns of clinical adoption. Finally, we developed a predictive model, RoboTRAC, to estimate translational probability and graded clinical potential from patent-level features, enabling identification of technology pathways most strongly associated with regulatory approval.

Findings: Our study provides a quantitative predictional tool (Robotic Translational Assessment for Clinical Impact, RoboTRAC) for anticipating clinically viable robotic technologies and offers engineering-relevant guidance for designing surgical systems with higher translational potential.

Interpretation: Global investment in surgical robotics has increased substantially in recent years. However, the efficiency of these investments and the extent to which research and development efforts are translated into clinically approved surgical robotic systems remain insufficiently characterized. In addition, systematic global analyses examining the progression of surgical robotic technologies from intellectual property development to clinical application are currently limited.

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INTRODUCTION

Surgical robot systems have performed millions of surgeries worldwide¹. More than 4,000 robots are used in clinical practice every day². In 2024, the global market size of surgical robots will exceed \$10 billion and maintain a compound annual growth rate of over 20%³. Due to the high density of technological innovation, the difficulty of clinical application, and the high legal regulatory requirements of surgical robots, competition is significantly higher than in the general medical device field. For example, after the early core patents of Intuitive Surgical expired, the market entry threshold was significantly lowered, and various companies and medical teams accelerated their entry, forming a large number of derivative technology paths around the da Vinci system⁴. However, in this process, is there a large amount of wasted funds due to repeated

investment? Can many competing companies transform their R&D technologies into clinical products?

Investigating the correlation between technology patents and their respective clinical products is fundamental to comprehending technological and application development⁵⁻⁷. Currently, several challenges persist. First, significant disparities exist in the advancement of surgical robotics across technology classifications, capital outlays, and regional placements, leading to a deficiency in worldwide strategic analysis. In the United States, the adoption rate of robotic surgery is about 5% (healthcare expenditure accounts for 18% of GDP); in contrast, the adoption rate in the European Union is only about 2% (healthcare expenditure accounts for 10% of GDP), while in most other parts of the world is still below 1%⁸. Secondly, issues like extended R&D periods and significant capital expenditure indicates in an absence of clear guidance for industry planning. Third, a large number of patents have not entered the clinical application stage, forming a considerable number of “silent patents”, impiled a lack of standards for evaluating patent quality. For example, the number of engineering papers related to therapeutic rehabilitation and assistive wearable robots is 32 times that of medical papers⁹. Traditional patent is difficult to answer which technology patents can pass regulatory review and enter the real medical environment^{10,11}. Therefore, linking technology patents with clinical products can effectively identify high-conversion technologies and optimize the allocation of industrial resources.

Research in context

Evidence before this study

On Mar,10, 2026, we searched PubMed for articles published between Jan 1, 2020, and Mar 10, 2026, without language restrictions. The search terms included (“robot*” OR “surgical robot” OR “surgery robot” OR “clinical robot”) AND “patent”. Existing studies primarily only examined patent activity in clinical robotics focusing on filing trends, geographic distribution, and broad technological categories. A study published in 2025 reported that collaborative robot technologies are entering a more mature stage, characterized by the dominance of invention patents, a concentration of innovation activity in the USA and China, and leadership from both multinational corporations and emerging enterprises. However, the available literature has not systematically examined how the observed patent landscape relates to technological positioning within surgical robotics or how patent distributions may inform the future development of next-generation surgical robotic systems.

Added value of this study

Our study expands on analyzing the relation between surgical robot's technology patent and clinical products. As the same aim of previous study, this study focused on a global technology innovation time trends. We systematically examine patent data across technological domains and geographic regions, analysis its driven clinical product. Building on the existing descriptive patent literature, this study provides a descriptive analysis of how the surgical robot technology has been integrated into clinic field. Moreover, this study analyzed the patents with the greatest clinical commercialization potential, providing a quantitative predictional tool (RoboTRAC) which can further predict whether a new patent have great clinical usage potential before it steps into real application period. Our study finds shows that only less than 30% of the surgical robot patents has developed into clinical application, less than 7% of the patent technology has great potential in clinical usage. This reveals that only a few technological trajectories account for the majority of downstream medical device approvals. Currently, all the surgical robots integrated into clinic usage concentrated on minimally invasive procedures (as shown in Fig.1). Our results highlight where the largest disparities are in robot distribution which is lead by a few top corporations (such as Covidien LP and Intuitive Surgical Operation, Inc) and few major countries such as America and China (as shown in Fig.2 & Fig.3). By applying large language model, we create a tool to future analysis a patent's clinical commercialization's potential based on patent's description, this avoid further funding waste and unnecessary resource usage.

Implications of all the available evidence

By contextualizing technology's real application distributions, this study extends prior work from simple trend reporting to a more structured interpretation of how intellectual property may reflect and shape future innovation trajectories. Our research indicates that substantial resources were squandered during the patent development phase due to a weak link to clinical use. Our research introduces a novel framework for determining whether a patent includes enough information for functional implementation that could avoid funding waste. Moreover, our study reflects that currently exist a large extend of surgical robot disparity, there is scope for all countries and new corporations to scale up the development of surgical robot Patent landscape analyses can offer valuable insights into the global trends, technological priorities, and emerging directions in surgical robotics. A more structured understanding of these patterns may support researchers,

industry stakeholders, and policymakers in assessing innovation dynamics and anticipating areas of future technological development in surgical robotic systems.

To overcome the disconnect between technological development and clinical translation, we directly link technology to clinical medical devices by combining global surgical robot patent data with regulatory approval records from major regulatory agencies, including the U.S. Food and Drug Administration (FDA) and the China National Medical Products Administration (NMPA). By reconstructing the analysis process around clinical product translation, this study answers a core question: What are the current characteristics of the global surgical robot patent landscape? And which technological directions are most likely to generate clinical applications in the future? How can we predict the quality of future new patents? This is crucial for reducing development costs, providing strategic planning, and ultimately improving the quality of clinical surgery¹².

MATERIALS AND METHODS

Study design

Through applicant-based linkage and relevance filtering, we identified 9,584 products that could be reliably associated with specific patent portfolios. Using applicant-level linkage, manual and automated relevance labeling, and supervised machine learning, we quantify translational likelihood and stratify patents by predicted clinical conversion potential.

Participants

Researcher retrieved patents related to surgical robotic systems from the PatSnap Global Patent Database, covering filings between January 1, 2005 and December 31, 2024 from 102 countries (172 patent offices) worldwide.

Patent retrieval was performed using a composite Boolean query combining surgery-related procedural action terms, robotic system and module descriptors, and IPC-based inclusion and exclusion filters. Both Chinese and English keywords were included to capture domestic and international filings. The full search expression is provided in the Supplementary Methods.

Automated and manual denoising process was applied. After denoising, 36,686 patents were retained for downstream analysis.

Procedures

Assignee-based bridge construction and product pool assembly

Researcher extracted assignee names from the surgical-robotics patent corpus and used assignee identity as the primary bridge to retrieve candidate regulatory products. Assignee names were standardized to consolidate obvious variants (e.g., legal suffixes and language forms). Using the resulting assignee list, researcher queried FDA and NMPA device databases to collect all regulatory-cleared products registered by these entities between 2005 and 2024.

Product relevance screening

Researcher screened the assignee-linked product pool to remove approvals unrelated to surgical robotics. To establish inclusion and exclusion criteria, researcher manually labeled 7,418 product records or candidate patent–product instances based on intended clinical use and device type. Using this manually labeled subset for calibration, researcher applied DeepSeek-assisted semantic screening to the remaining 30,016 products to identify records plausibly related to surgical robotics within the study scope. After screening, 9,584 products were retained as surgical-robotics–relevant and eligible for patent linkage. Large language model (LLM) outputs were used only to support large-scale screening and consistency checking, not for final adjudication.

Feature construction

Each patent was represented using structured patent attributes and semantic embeddings. Structured features included patent age, title length, abstract length, number of claims, patent type, and categorical encodings of IPC and CPC classifications. To capture functional and architectural similarity beyond classification codes, concatenated patent titles and abstracts were encoded using a BGE sentence embedding model. Embedding vectors were normalized and reduced using principal component analysis (PCA) to retain dominant semantic dimensions while reducing noise. Structured and embedding-based features were concatenated to form final model inputs.

Robotic Translational Assessment for Clinical Impact Tool (RoboTRAC)

Two supervised prediction tasks were formulated.

Binary classification

Patents were labeled as translated or non-translated based on patent–product linkage. A LightGBM gradient boosting classifier was trained using stratified 80/20 train–test splits.

Four-class ordinal classification

For graded translational assessment, patents were labeled as Q1–Q4 based on quartile stratification of matched product counts. A multi-class LightGBM classifier was trained using class-balanced sampling.

RESULTS

Study Overview

We integrated global patent data with regulatory approvals from FDA and NMPA to examine how surgical robotic technologies progress from invention to regulatory-cleared clinical systems between 2005 and 2024 (Fig. 1).

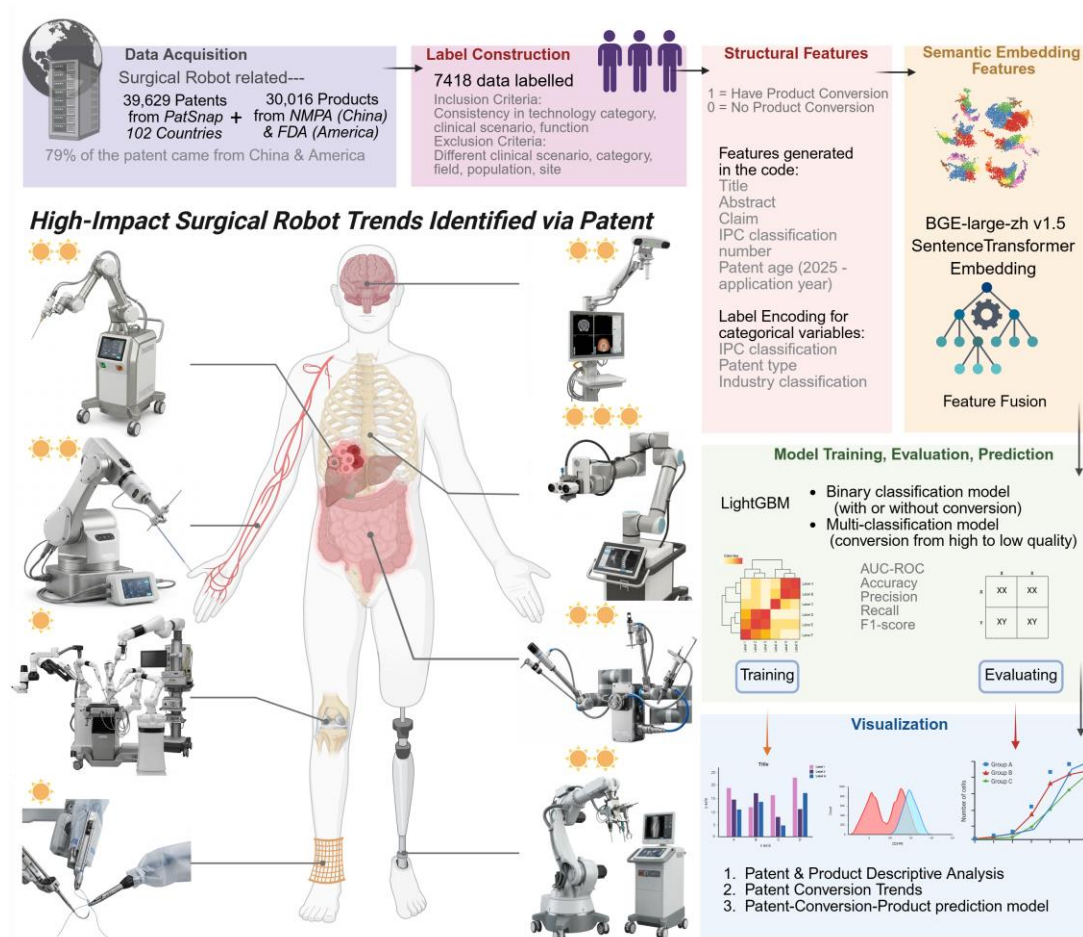


Fig. 1. Patent-Conversion-Product Trend Analysis and Prediction Protocol. The sunburst surrounding the central schematic represents the overall level of technological activity across major domains of surgical robotics. From the upper left to the lower right, the sectors correspond to: Percutaneous Ablation Robotics, Interventional Catheter Navigation Systems, Arthroscopic Surgical Robotics, Robotic Suturing Systems, Stereotactic Neurosurgical Systems, Orthopedic Navigation Systems, Minimally Invasive Surgical Platforms, and Fracture Reduction Robotics.

Global Patent Landscape of Surgical Robotics (2005–2024)

Invention patents constitute the vast majority of all patents, while utility model and design patents account for only a small proportion (Fig 2A). Meanwhile, the high percentage of valid patents indicates that most related technologies remain under active protection (Fig 2B).

Patent value exhibits a clear gradient distribution, with 60.6% of patents concentrated in the range of 10,000 to 30,000 yuan (Fig 2D), and patents valued between 30,000 and 300,000 accounting for 24.8%. These relatively low-value patents continue to play a crucial supporting role in the ongoing development and application of surgical robot technology.

The patent application trends of major applicants clearly outline the evolution of the industry's competitive landscape (Fig 2C and 2E). Industry leaders such as Covidien LP experienced explosive growth between 2015 and 2022, maintaining exceeding 100 from 2014 to 2022. Intuitive Surgical Operation, Inc exhibited sustained growth, achieving over 100 applicants as early as 2006. Concurrently, Olympus Corporation displayed considerable instability. Ethicon Endo-Surgery, Inc.'s figures fell to zero in subsequent years.

The core concepts of surgical robot patent technology are summarized, with successive outer layers referring to system-level hardware, key mechanical components, and procedure-specific applications (Fig .3A). Intermediate categories include surgical systems and surgical equipment, whereas component-level branches comprise robotic arms and actuators. Application-oriented branches extend into specific procedural domains, including orthopedic surgery. Outer-ring terms represent finer-grained subdivisions of these categories, indicating increasing technical specialization built upon shared tool-based and system-level architectures. The worth of diverse patents can be found in Fig. 3B.

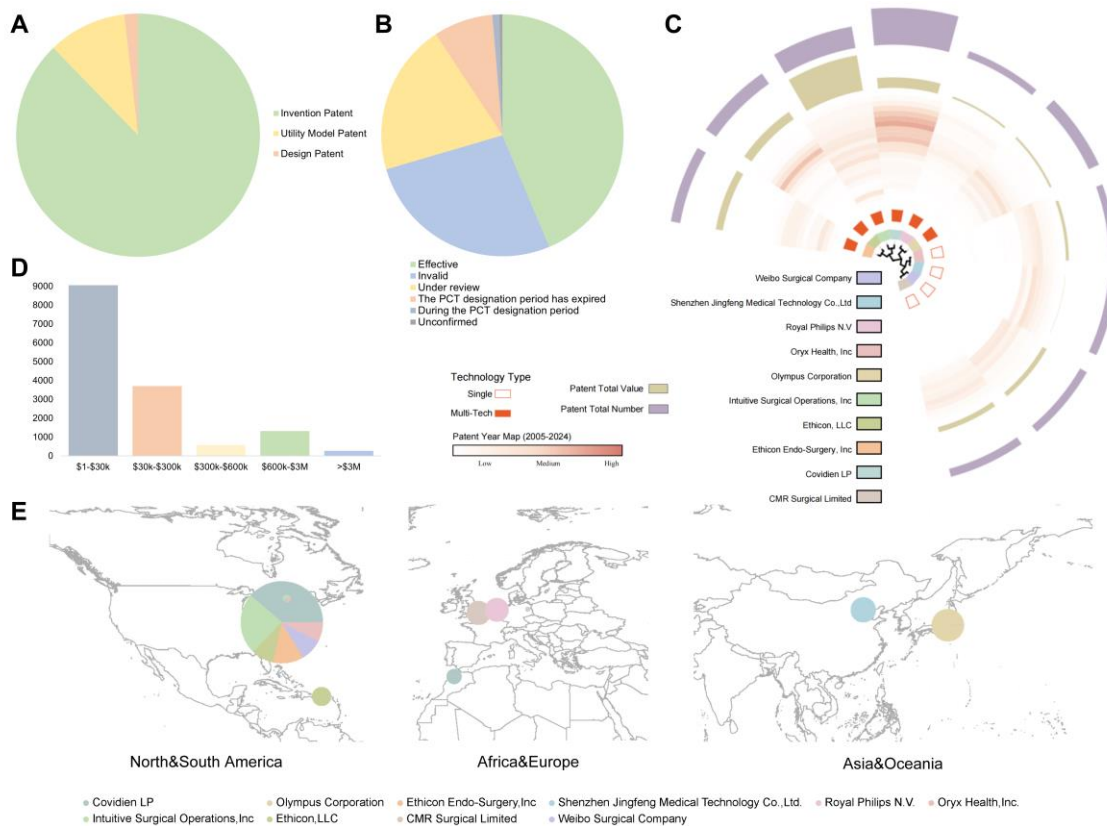
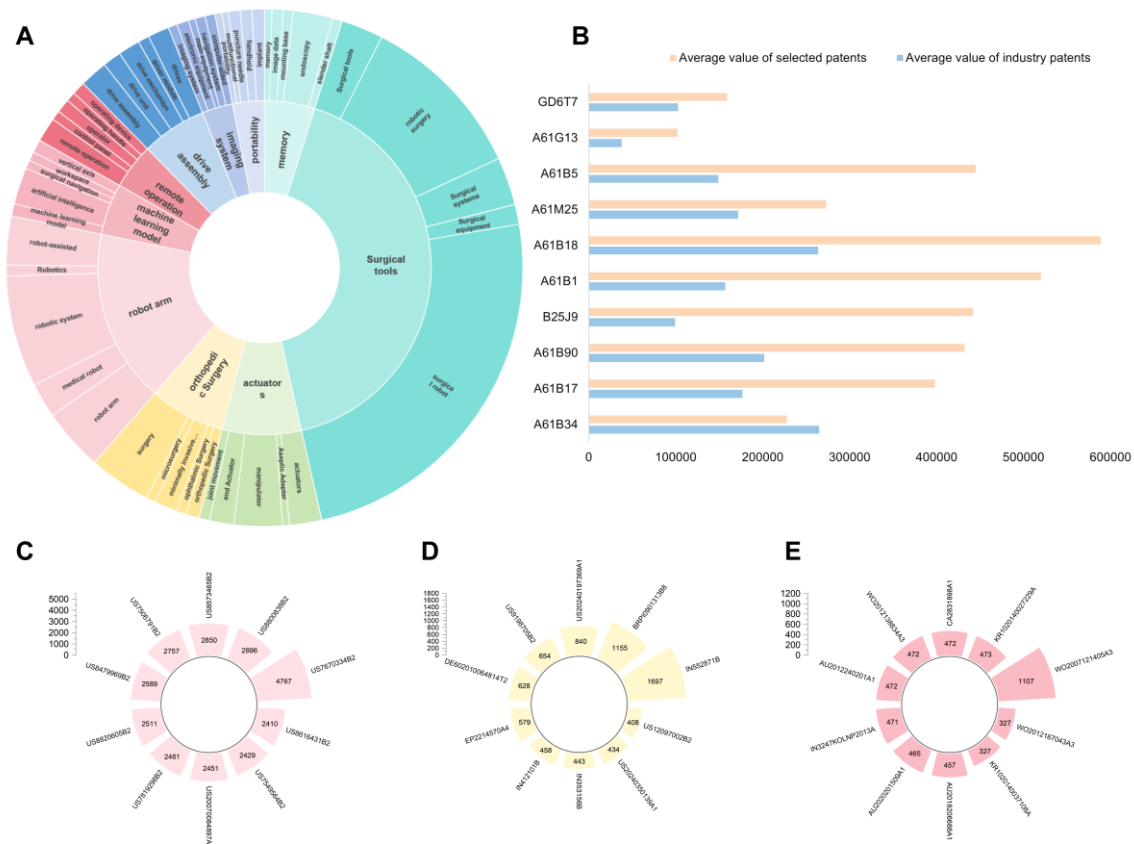


Fig. 2. Patent characteristics and global distribution of major applicants in surgical robotics. (A)

Distribution of patent types, including invention, utility model, and design patents. **(B)** Legal status of patents, categorized as effective, invalid, under review, within or beyond the PCT designation period, and unconfirmed. **(C)** Radial timeline of patent application activity among major applicants, illustrating longitudinal filing patterns across companies. **(D)** Distribution of estimated patent value across predefined monetary intervals. **(E)** Geographic distribution of major applicants, with pie charts indicating regional patent portfolios in North and South America, Africa and Europe, and Asia and Oceania.

Highly cited patents in surgical robotics are overwhelmingly concentrated on core system components and surgical instruments, reflecting the central role of hardware architecture and end-effector design in foundational innovation (Fig. 3C). These patents primarily focus on system-level core components and are all originating from the United States. Eight of them were filed by Ethicon Endo-Surgery, Inc., demonstrating a clear institutional concentration. US7670334B2 is the most cited patent; its proposed articulated surgical instrument structure improves instrument accessibility within the surgical field, and the multi-angle adjustment capability of the end effector helps support minimally invasive procedures.

Patent family size is used to measure the strength of a patent holder's global footprint (Fig. 3D). The largest patent family is IN552871B (Ethicon, LLC), containing 1,697 patents across 11 jurisdictions, including the United States, China, Japan, and India. These jurisdictions represent major medical device markets. It indicating extensive international filing coverage for this invention.



with the highest number of claims.

Divergence Between Patent Growth and Clinical Deployment

The gap between applicant patent and granted patents has widened in recent years, indicating declining authorization efficiency. The percentage of patent applications that were granted in the same year (the authorization rate) was relatively stable at about 60–70% before 2017 (Fig. 4A); however, it notably decreased in the following years, stabilizing around 20–30% past 2021. Growth in patent filings has not been matched by proportional growth in regulatory-cleared products, revealing increasing decoupling between technological development and clinical deployment. Despite a sustained increase in patent filings from 2005 to 2024, annual counts of regulatory-cleared products rise more slowly and with marked temporal fluctuations, indicating a growing decoupling between upstream technological exploration and downstream clinical deployment in surgical robotics (Fig. 4B).

Technological exploration remains geographically concentrated, with sustained dominance by the United States and delayed but rapid expansion in China. Since 2005, America has been at the forefront, while China only began to expand after 2015. Other countries have maintained a minor share over the last two decades. In the years 2023-2024, there was a decline in the total amount of technology discovered (Fig. 4C). Patterns of market entry differ from those of technology origin, with China showing continued growth in regulatory deployment after 2015, while U.S. market expansion appears comparatively attenuated (Fig. 4D).

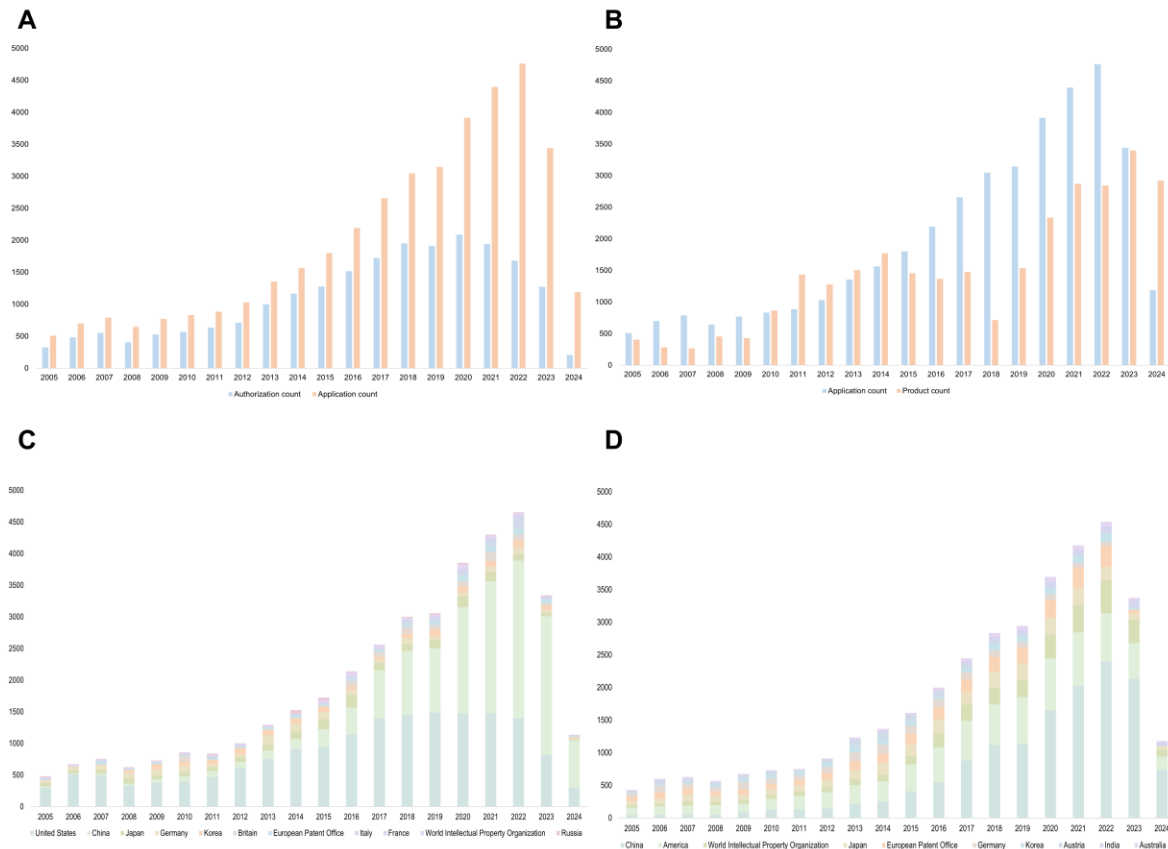


Fig. 4. Global landscape and temporal evolution of patents and commercial products related to surgical robotic systems (2005–2024). (A) Annual trend of applications and granted patents. (B) Yearly counts of surgical robotic products and associated patents. (C) Annual trend of geographical distribution of technology origin (source countries/regions). (D) Annual trend of geographical distribution of target markets (destination countries/regions).

Despite sustained growth in patenting activity, clinical deployment of surgical robotic technologies remains highly selective and structurally constrained, resulting in a pronounced mismatch between innovation volume and real-world clinical uptake.

Clinical deployment is highly concentrated within a small subset of technological pathways, despite comparable patent volumes across translational potential groups. Patents in the highest-potential group (Q1) are associated with substantially more matched regulatory products (37,010) than those in lower groups (Q2: 18,395; Q3: 8,626; Q4: 4,323), indicating strong concentration of downstream deployment within a limited subset of technological pathways (Table 1). In contrast, average patent-to-product conversion times differ only modestly across quartiles (3.96–4.84 years), suggesting that

translational heterogeneity arises primarily from differences in deployment scale rather than speed of regulatory clearance.

Table. 1. The Patent Conversion details. Matched Count is the patent-product-conversion matching order of magnitude. For graded assessment of translational intensity, translated patents were further stratified into four ordered categories based on the strength of their patent–product linkage. Patents were ranked by the number of matched regulatory-cleared products, and quartile-based stratification was applied: top 25% as Q1 (high translational potential), 25th–50th percentile as Q2 (moderately high), 50th–75th percentile as Q3 (moderate), and bottom 25% as Q4 (low).

Conversion Potential	Patent Count	Average Conversion Time (year)	Matched Count
Q1 (High)	2259	4.83826	37010
Q2 (Moderately High)	2259	4.269505	18395
Q3 (Moderate)	2258	4.015542	8626
Q4 (Low)	2259	3.955562	4323

Surgical Robot Technology: Clinical Application Landscape

Patents referring to general robotic systems and manipulation platforms account for the largest share. Over time, multi-modal and intelligent enhancement components have been gradually incorporated (Fig. 5A).

In parallel, patents involving endoscopic coupling and image-guided operation, increase progressively, indicating a expanding of auxiliary clinical technologies into robotic systems. Visible expansion of control systems powered by patented computer learning has been observed in more recent times, coinciding with the adoption of data-driven surgical technologies.

Surgical robot technology advancements focused on enhancing basic targeting precision. Technology that related to accurate lesion localization and spatial targeting technology remain consistently prominent (Fig. 5B). Patents addressing control precision and operational stability increase steadily. In contrast, safety-oriented objectives, including prevention of secondary injury and reduction of cross-contamination, appear less frequently in early periods but become increasingly represented in later years, reflecting rising emphasis on procedural safety and perioperative risk control.

However, technological prevalence in patent space does not necessarily imply clinical deployment. To determine which of these innovation directions are selectively retained through regulatory and clinical filtering, we next examined how different technical themes translate into approved products and patent–product linkages (Fig. 5C and 5D). This transition from patent-level evolution to product-level realization reveals which technological trajectories persist beyond invention and which remain confined to exploratory development. Topic modelling of patents linked to regulatory-cleared products identified 15 technological themes spanning robotic structures, control and sensing, navigation, interventional instruments, and procedure-specific system designs (Table 2).

Table 2. Product Theme Topics.

Theme ID	Keywords
1	rotation, linkage, axis, platform, support, connection, direction, center, manipulator, joint, translation, motion, base, robot, articulation
2	control, sensor, signal, input, controller, detection, operation, ultrasound, handle, reception, output, measurement, probe, actuator, position
3	laparoscope, endoscope, surgical instruments, articulation, multiple degrees of freedom, control, suturing, stapler, puncture, tissue, cutting, tools, minimally invasive, joint, visualization
4	spine, fracture, fixation, navigation, positioning, guidance, stereotactic, orthopedics, structure, channel, driving, tools, components, mechanism, operation
5	catheter, puncture, vessel, navigation, guidance, positioning, visualization, imaging, ultrasound, tissue, guiding cannula, channel, insertion, manipulation, interventional procedure
6	drive, components, rotation, parts, transmission, catheter, connection, gear, power, motor, transfer, operation, housing, direction, motion
7	tissue, surface, insertion, guidance, therapy, cutting, structure, implant, region, spine, material, electrode, opening, implantation, flexibility
8	suturing, stapler, tissue, surgical instruments, articulation, flexibility, tools, operation, laparoscope, puncture, cutting, visualization, guidance, control, manipulator
9	elements, design, pulley, shape, application, photo, image, name, stereogram, interface, variation, clamp, minimally invasive, surgical operation
10	stereotactic, navigation, positioning, guidance, puncture, channel, visualization, imaging, tissue, catheter, ultrasound, structure, operation, drive, delivery
11	puncture, therapy, ultrasound, visualization, imaging, guidance, positioning, tissue, channel, ablation-related, operation, catheter, cannula, control, imaging
12	joint, orthopedics, fracture, cutting, tissue, visualization, tools, channel, fixation, drive, operation,

	manipulator, support, anatomy, guidance
13	fracture, fixation, reduction, guidance, positioning, support, channel, tools, tissue, structure, drive, operation, orthopedics, visualization, connection
14	instruments, unit, surgical, sterile, interface, surgical instruments, device, component, driver, motor, adapter, connection, surgical operation, coupling
15	configuration, robot, structure, position, connection, coupling, support, extension, positioning, frame, body, housing, reception, surface, cart

Product approvals are highly concentrated in a small number of themes (Fig. 5C). Among them, the eight surgical robot fields with significantly high transformation degrees are respectively Orthopedic Navigation Systems, Minimally Invasive Surgical Platforms, Interventional Catheter Navigation Systems, Fracture Reduction Robotics, Stereotactic Neurosurgical Systems, Percutaneous Ablation Robotics, Arthroscopic Surgical Robotics, and Robotic Suturing Systems (shown in Fig. 1). These themes predominantly correspond to task-specific operative tools, navigation modules, and procedure-level robotic assistance rather than to general robotic platforms.

A similar concentration is observed in patent–product matching counts (Fig. 5D), indicating that patents associated with surgical workflow integration are substantially more likely to correspond to approved devices. In contrast, themes dominated by mechanical structures, transmission components, or generic actuation technologies contribute large numbers of patents but rarely translate into standalone clinical products.

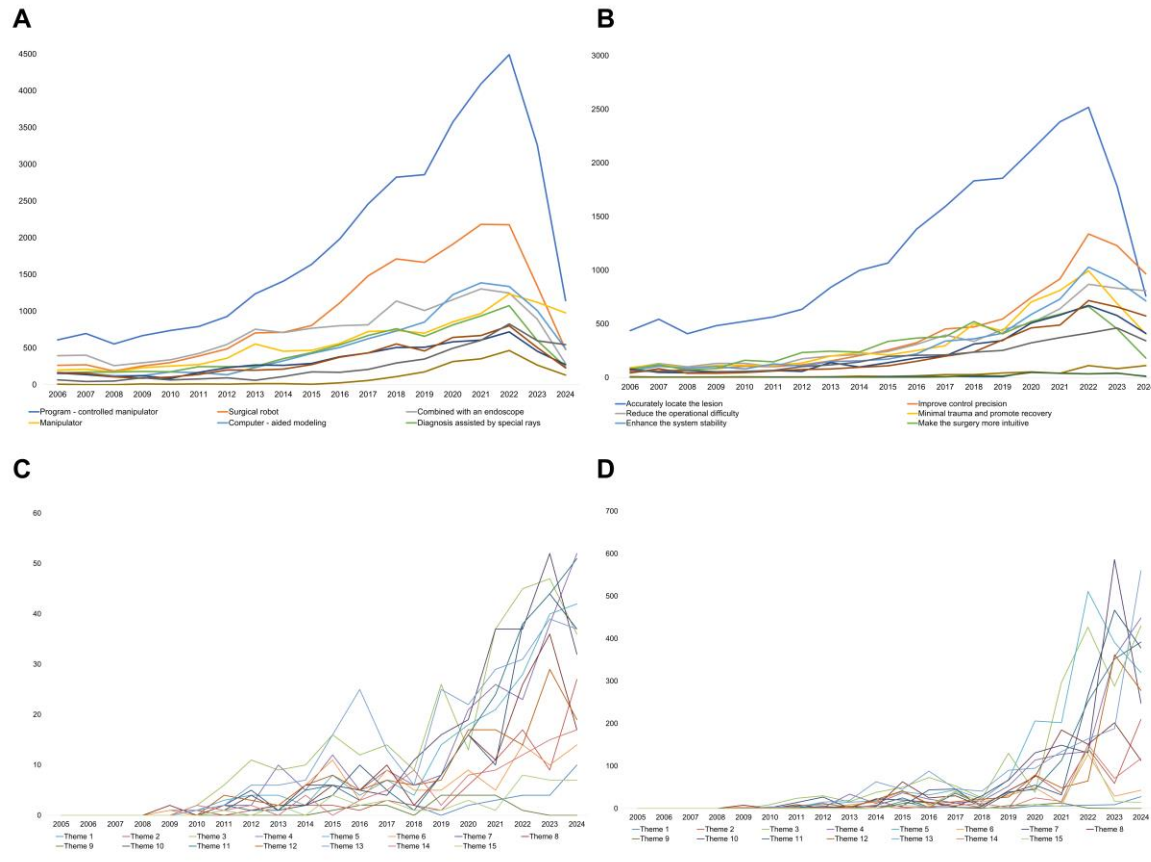


Fig. 5. Innovation characteristics and thematic evolution in the global surgical robotics patent landscape. (A) Distribution of technical categories. (B) Technical efficacy of patented innovations. (C) Distribution of converted products across technological themes identified by topic modeling. (D) Distribution of patent–product matching counts across technological themes.

Future Clinical Translation and Predictable Technology Tool: RoboTRAC

Machine learning models were able to identify patents with high conversion potential. The binary classification model achieved high precision and recall (90% and 89%, respectively) in the "unconverted" category, indicating that the model can reliably identify patents unlikely to enter clinical application; however, its performance in the "converted" category was moderate (68% precision, 71% recall), reflecting the high heterogeneity inherent in successful conversion pathways (Table 3).

In the multi-class classification model, the differentiation between different quartiles of conversion potential showed moderate predictive accuracy, with AUCs for each category ranging from 0.73 to 0.84. The Q1 group, with the highest conversion potential, had an F1 score of 0.61 and an AUC of 0.84, suggesting that patents most likely to achieve product conversion could be identified with relatively reliable confidence.

Table. 3. Performance of binary and four-class classification models for predicting patent–product translation.

Classification	Precision	Recall	F1-Score	AUC-ROC
<i>Binary Classifier</i>				
No Product Conversion	90%	89%	90%	-
Have Product Conversion	68%	71%	70%	-
<i>Multi-Class Classification Model</i>				
Q1 (High)	0.5564	0.6785	0.6114	0.8390
Q2 (Moderately High)	0.5601	-	0.5018	0.7490
Q3 (Moderate)	-	-	-	0.7287
Q4 (Low)	0.5221	0.5233	0.5227	0.7844

DISCUSSION

Nowadays, surgical robot still lacks a suitable standardized review system. Past instances demonstrate that the traditional regulatory frameworks, including the FDA, have not evolved alongside the increasing capabilities (such as autonomy) of surgical robots, which has impeded their clinical application progress¹³. To solve this problem, Nature published a standard in evaluating surgical robot’s R&D cycle and divided into four stages: preclinical development and early clinical evaluation, comparative evaluation, long-term monitoring and technological development¹⁴. Various nations have implemented distinct approaches, with Brazil, for instance, enforcing CFM Resolution 2,311/20226, which mandates the presence of at least one human physician during robotic surgeries¹⁵. Nonetheless, this failed to stop the inefficient spending on similar technology in the beginning. Consequently, the importance of safeguarding patents cannot be overlooked.

Considering the established assessment system alongside the economic advancement status of various regions, the global distribution of surgical robotic technology is highly imbalanced. Countries with advanced economies, like the United States, Europe, and Japan, possess the

majority of surgical robotic systems⁸. Surgical robots increased the precision and safety of healthcare services^{16,17}. For instance, the Da Vinci surgical robot facilitates the advancement of minimally invasive surgical techniques. It does not only reduce recovery times for patients, but also increase surgical success rates. Meanwhile, it minimized complications and the need for referrals due to insufficient medical conditions¹⁸. Furthermore, the emergence of remote techniques could help mitigate the lack of expert resources in less developed regions¹⁹. Nevertheless, several things such as the costs of equipment, the state of network infrastructure, and the training of local medical personnel must keep on improving²⁰.

At this time, robotic surgery is employed across clinical fields²¹ such as orthopedics, urology, general surgery, and obstetrics. From patients' viewpoint, robot-assisted procedures result in smaller incisions, decreased blood loss, and faster healing times²². Such benefits could be seen in breast reconstruction, tumor excision with minimal invasiveness, joint replacements, and brain biopsies^{19,23}. From doctor's viewpoint, surgical robots enable doctors to adopt a more comfortable sitting arrangement, minimizing fatigue and boosting their concentration levels²⁴. Moreover, surgical robots have achieved higher rates of success and lower instances of complications in precise operations such as microvascular suturing and addressing lymphedema. In fields such as neurosurgery, spinal, pediatric, cardiothoracic surgery, the use of robotics is still in the early stages of development, indicating a considerable potential for progress¹⁹. The aforementioned fields face unique anatomical and engineering challenges that existing robotic systems are still struggling to address effectively. This suggesting significant potential for technological evolution and application expansion in clinical settings²¹.

The development of surgical robot is strongly related to economic support. Based on our patent review, the patent majority hold a value between \$1,000 and \$30,000, which could potentially allow the company to reach its break-even point. Olympus Corporation holds a market capitalization of \$16,844,936,629, with 8.62% allocated to research and development. If patent protections were enhanced, we could foresee an increase in R&D investment. For instance, the OPERATIONS INSTRUMENT MIT EINEM MEHRSCICHTIGEN ANTRIEBSSTRAHL from Covidien LP is valued at \$10,500,000. As expected, once it expires in 2027, the associated technology is projected to experience substantial growth.

As outlined in the patent application, the surgical robot has seen multiple alterations aimed at maximizing its efficiency. The first generation enabled stereotactic procedures on rigid tissues under full human control. The second generation introduced systems like Da Vinci²⁵, enhancing

minimally invasive soft tissue surgery via 3D imaging and articulated instruments. Third-generation bionic robots, such as i-SNAKE and Niobe, navigated complex anatomies like vessels using snake-like or magnetically guided designs. The fourth generation focused on micro-robots, including capsule endoscopes, now advancing toward active navigation and therapeutic use. Fifth-generation systems aim for full autonomy using AI and swarm intelligence, though true cognitive independence remains unrealized. In summary, surgical robots have shifted from mechanical aids to intelligent, miniaturized tools. Future developments emphasize autonomy, bio-inspiration, and targeted intervention, though major scientific breakthroughs are still needed for fully autonomous clinical applications²⁶.

The progression of techniques has moved from mechanical support to intelligent systems, with every enhancement aimed at fulfilling clinical requirements.

As advancements in technologies like AI-powered automation, nanotechnology in robotics, less invasive surgical techniques, semi-automated robotic systems, and the influence of 5G connectivity on remote surgery continue to emerge²⁷, future advancements are geared towards smaller designs²⁸, nature-inspired systems, and independence, although the creation of entirely autonomous technologies still relies on key scientific discoveries.

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Supervision: KS, SZ, XC, Writing – original draft: KH, JD, HC, YX, Writing – review & editing: ZL, XX, JX

Conflict of Interest Disclosures: Authors declare that they have no competing interests.

AI declaration statement: the study only use AI to create surgical robots schematic diagram section showing in Fig.1 to avoid possible infringement.

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